

MindTrack AI: Intelligent Digital Behavior & Social Media Addiction Analyzer

AI23331 - FUNDAMENTAL OF MACHINE LEARNING

Submitted by

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BONAFIDE CERTIFICATE

Certified that this Project titled “**Hybrid Handwritten Text Recognition System using OCR and Transformer-Based Deep Learning**” is the bonafide work of “**MOHANA PRIYA R (230701191), PRATHEBA (230701239)**” who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

MindTrack AI is an intelligent web-based application developed to analyse and monitor users' digital behaviour, with the goal of identifying patterns that may indicate social media overuse or digital addiction. The system continuously observes key usage metrics such as daily screen time, frequency of device unlocks, time spent on social media applications, and late-night usage patterns. Using these inputs, it applies a rule-based simulation model to generate a Digital Addiction Risk Score ranging from low to high.

Based on the calculated score, the application classifies user behaviour into categories such as normal, moderate, or high risk. It provides real-time insights through visual dashboards, including charts and dynamic statistics, allowing users to better understand their usage habits. The system also generates personalized recommendations, such as reducing screen time, taking breaks, or enabling focus sessions, to promote healthier digital routines.

A key feature of **MindTrack AI** is its interactive chatbot, which responds to user queries, provides usage summaries, and suggests actionable steps to improve digital well-being. Additionally, the application includes a real-time alert system that notifies users when excessive usage or unhealthy patterns, such as prolonged screen time or late-night activity, are detected.

Although the system uses simulated logic instead of a fully integrated machine learning model, it effectively mimics AI-driven behaviour, making it suitable for demonstration and practical understanding. Overall, **MindTrack AI** aims to enhance user awareness, improve productivity, and encourage balanced technology usage through an engaging and user-friendly interface.

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LIST OF ABBREVIATION

1.	AI	Artificial Intelligence
2.	ML	Machine Learning
3.	DARS	Digital Addiction Risk Score
4.	UI	User Interface
5.	UX	User Experience
6.	HCI	Human-Computer Interaction
7.	DA	Data Analytics
8.	RTAS	Real-Time Alert System
9.	HTML	HyperText Markup Language
10.	CSS	Cascading Style Sheets
11.	JS	JavaScript
12.	DB	Database
13.	API	Application Programming Interface
14.	CNN	Convolutional Neural Network
15.	RNN	Recurrent Neural Network
16.	LSTM	Long Short-Term Memory
17.	RF	Random Forest
18.	LR	Logistic Regression
19.	GB	Gradient Boosting
20.	OCR	Optical Character Recognition
21.	NLP	Natural Language Processing
22.	SQL	Structured Query Language
23.	UI/UX	User Interface and User Experience
24.	JSON	JavaScript Object Notation
25.	HTTP	HyperText Transfer Protocol

CHAPTER 1

INTRODUCTION

1.1 GENERAL

In today's digital era, the widespread use of smartphones and social media platforms has significantly increased screen time and online engagement. While technology enhances connectivity and productivity, excessive usage often leads to negative impacts such as reduced concentration, poor sleep quality, decreased productivity, and mental health concerns. Many users are unaware of their actual usage patterns and the long-term effects of digital overconsumption.

MindTrack AI is developed as a smart web-based solution to address this issue by analyzing user digital behavior and identifying potential signs of addiction. The system monitors key parameters such as screen time, app usage frequency, unlock count, and night-time activity. Based on these inputs, it generates a **Digital Addiction Risk Score** and provides insights into user habits.

0 The application also includes features such as real-time analytics, personalized recommendations, alert notifications, and an interactive chatbot. Although it uses rule-based logic instead of a fully integrated machine learning model, it effectively simulates intelligent behavior, making it suitable for demonstration and practical implementation. The system dynamically updates user data and generates insights, helping users clearly understand their usage patterns. The overall goal is to promote digital awareness and encourage healthier technology usage.

1.2 OBJECTIVE

- To analyze and monitor user digital behavior using key usage metrics
The system tracks parameters like screen time, app usage, and unlock count. This helps in understanding user interaction patterns effectively.
- To detect patterns that indicate excessive usage using AI-based concepts
The application identifies unusual and repetitive usage behaviors. It uses AI-based logic to detect possible addiction patterns.
- To generate a Digital Addiction Risk Score and classify users into different levels
A score is calculated based on user activity data. Users are categorized into low, moderate, or high-risk levels.
- To provide real-time insights and visualizations for better understanding
Data is displayed using charts and dynamic statistics. This helps users easily understand their usage behaviour.
- To offer personalized recommendations for reducing screen time
The system suggests actions based on user behaviour. These help in improving digital habits and reducing overuse.
- To develop a chatbot that assists users with queries and suggestions
A chatbot provides instant responses to user queries. It improves engagement and offers helpful guidance.
- To implement a real-time alert system for unhealthy usage patterns
Alerts are generated when excessive usage is detected. This helps users take immediate corrective actions. To simulate ML-based predictions using rule-based logic for demonstration. The system uses predefined logic to mimic ML predictions. This simplifies implementation while demonstrating intelligent behaviour.

EXISTING SYSTEM

Currently, existing systems such as built-in digital wellbeing tools and mobile applications provide basic tracking of user activity, including screen time, app usage, and notification frequency. These systems mainly present data through simple **User Interfaces (UI)** using dashboards and reports, but they lack deeper analysis and interpretation. Most tools operate as **Monitoring Systems (MS)** that only display usage statistics without applying **Artificial Intelligence (AI)** or **Machine Learning (ML)** techniques to understand user behavior patterns.

Additionally, these systems do not include advanced features such as predictive analysis, personalized recommendations, or intelligent assistance. There is minimal use of **Data Analytics (DA)** to derive meaningful insights, and users are not provided with actionable feedback to improve their habits. Features like interactive chatbots, which could enhance **Human-Computer Interaction (HCI)**, are generally absent. Furthermore, existing systems lack an efficient **Real-Time Alert System (RTAS)** to notify users about excessive or unhealthy usage patterns.

Due to these limitations, current solutions fail to actively guide users toward healthier digital behavior and self-awareness. **MindTrack AI** addresses these gaps by integrating simulated AI-based logic, enhanced UI, real-time feedback mechanisms, and intelligent features into a single platform, thereby offering a more comprehensive and user-centric solution.

CHAPTER 2

LITERATURE SURVEY

“Digital Wellbeing and Smartphone Usage Monitoring System” [1] (2022) by Kumar R., et al. discusses a system for monitoring smartphone usage patterns and screen time to promote healthy digital habits. The study focuses on tracking user activity such as application usage frequency and device unlock count using mobile-based analytics. It highlights the importance of digital wellbeing tools in reducing excessive smartphone dependency. However, the system mainly provides basic monitoring features and lacks intelligent prediction or personalized recommendation mechanisms.

“Machine Learning Approaches for Social Media Addiction Detection” [2] (2023) by Sharma P., et al. explores the use of **Machine Learning (ML)** algorithms such as Logistic Regression and Random Forest for identifying social media addiction patterns. The research analyzes behavioral parameters including screen time, late-night activity, and social media interaction frequency to classify users into addiction levels. The study achieves promising prediction accuracy but requires large datasets and extensive model training for reliable performance.

“Real-Time User Behavior Analysis using Data Analytics” [3] (2021) by Lee J., et al. presents a real-time analytics framework for understanding user digital behavior. The system uses **Data Analytics (DA)** techniques to process user activity and visualize behavioral trends through dashboards and charts. The research demonstrates the effectiveness of real-time monitoring in identifying unhealthy usage habits. However, the framework lacks interactive assistance features such as chatbot integration and personalized alerts.

“AI-Based Recommendation Systems for Digital Wellness” [4] (2024) by Verma S., et al. introduces an intelligent recommendation system that provides personalized suggestions to reduce screen addiction and improve productivity. The model analyzes user activity patterns and generates recommendations such as break reminders and focus sessions. The study improves user engagement through adaptive suggestions, but the implementation complexity and dependency on real-time behavioral data remain significant challenges.

“Behavioral Pattern Detection using Deep Learning Techniques” [5] (2023) by Chen L., et al. investigates the use of deep learning models such as **Long Short-Term Memory (LSTM)** networks for detecting behavioral trends in social media usage. The research focuses on identifying patterns like doomscrolling, excessive night usage, and emotional dependency. Although the system achieves high accuracy in sequence prediction, it requires high computational resources and large-scale datasets, limiting its suitability for lightweight applications.

“Interactive Chatbot System for Mental Wellness Assistance” [6] (2022) by Ahmed T., et al. presents a chatbot-based system designed to provide users with guidance and support related to digital wellbeing and mental health. The chatbot uses predefined conversational logic and behavioral inputs to deliver personalized suggestions and reminders. The study highlights the importance of **Human-Computer Interaction (HCI)** in improving user engagement and accessibility. However, the chatbot lacks advanced contextual understanding and depends heavily on rule-based responses, limiting its adaptability in complex user interactions.

CHAPTER 3

PROPOSED SYSTEM

3.1 GENERAL

The proposed system, MindTrack AI, is an intelligent web-based application designed to analyze user digital behavior and detect potential addiction patterns. It uses concepts of AI and ML to process user activity such as screen time, app usage, and unlock frequency. The system applies DA techniques to interpret user behavior and generate meaningful insights.

The system provides real-time insights through an interactive UI and DARS. It also includes features like personalized recommendations, chatbot interaction, and a RTAS to notify users about unhealthy usage patterns. These features enhance UX and improve user engagement.

In addition, the system incorporates a lightweight BAM that continuously evaluates user activity and updates the risk score dynamically. This module ensures that users receive accurate and up-to-date feedback about their digital habits. The integration of multiple components enables smooth HCI and effective decision support.

Furthermore, the application is designed using modern web technologies such as HTML, CSS, and JS to ensure responsiveness and scalability. The system architecture follows a modular approach, allowing easy enhancement and future integration with real ML models or cloud-based services. Overall, the system aims to promote digital well-being and responsible technology usage.

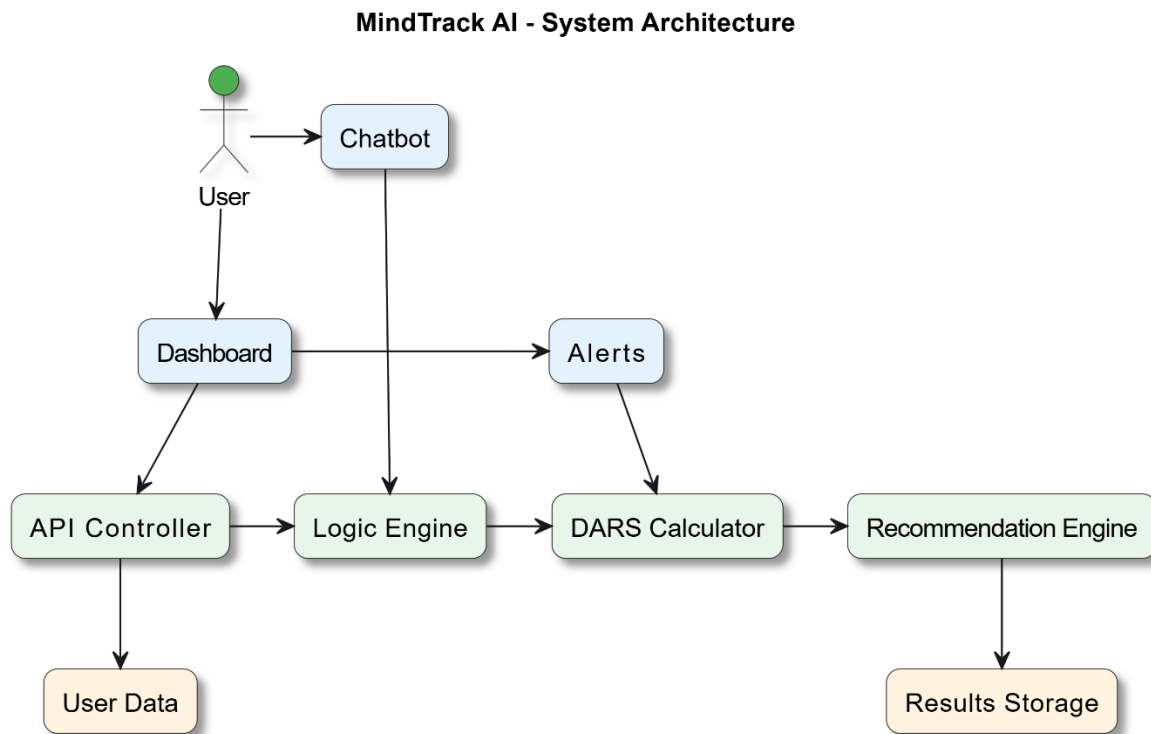
3.2 SYSTEM ARCHITECTURE

The system follows a simple architecture consisting of three main components:

1. **Frontend (UI Layer)** – Built using **HTML, CSS, JavaScript (JS)** for user interaction
2. **Processing Layer** – Handles logic, calculates DARS using rule-based simulation
3. **Data Layer** – Stores and manages user activity data

The user interacts with the UI, data is processed using logic, and results are displayed through dashboards, chatbot, and alerts.

Fig 3.1: System Architecture



3.3 DEVELOPMENTAL ENVIRONMENT

3.3.1 HARDWARE REQUIREMENTS

- System: Laptop/Desktop
- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Storage: 500 GB or above
- Internet Connection

3.3.2 SOFTWARE REQUIREMENTS

- Operating System: Windows / Linux / macOS
- Frontend: HTML, CSS, JavaScript (JS)
- Backend (optional): Python / Node.js
- Browser: Google Chrome / Edge
- Tools: VS Code

3.4 DESIGN OF THE ENTIRE SYSTEM

3.4.1 ACTIVITY DIAGRAM

The activity flow starts when the user opens the application. After accessing the system, the user inputs usage data such as screen time, app usage, and unlock count. Next, the system processes the input data and performs necessary computations. Based on this processed data, it calculates the Digital Addiction Risk Score (DARS) using predefined logic.

After calculating the score, the system checks whether the risk level is high or not. If the risk is high, the system generates an alert notification to warn the user. If the risk is not high, the system continues with normal monitoring of user activity.

Finally, the system displays the results including the risk score and insights through the interface. It also provides recommendations to help the user improve their digital habits, and the process ends.

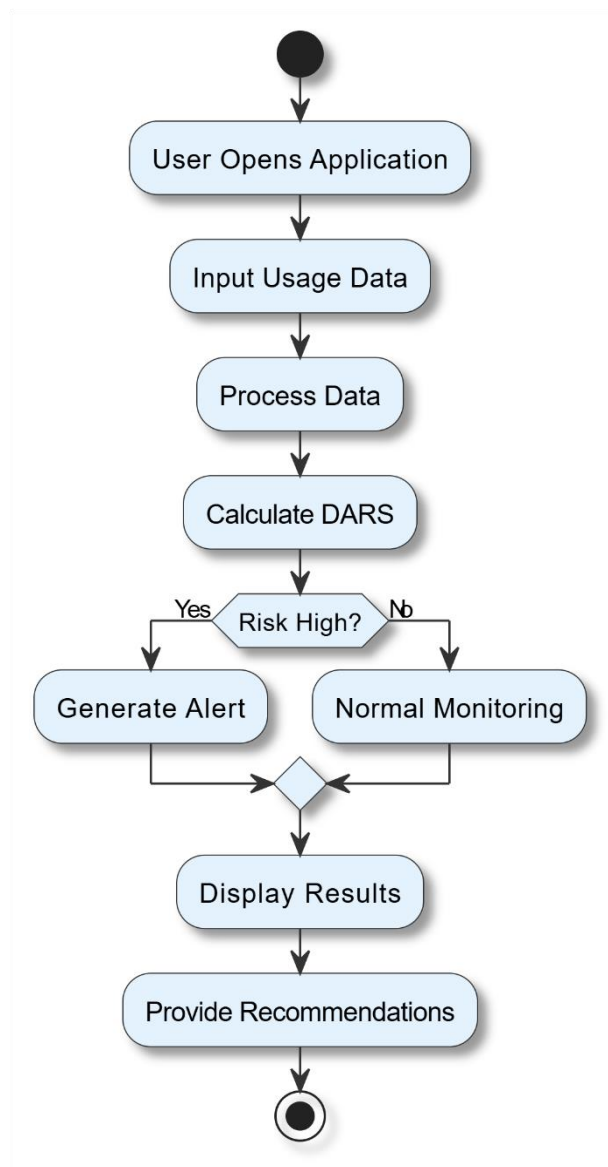


Fig 3.2: Activity Diagram

3.4.2 DATA FLOW DIAGRAM

The data flow begins when the **user provides input data** such as screen time, app usage, and unlock count to the system. This input is first sent to the **data collection module**, where all user activity data is gathered.

Next, the collected data moves to the **processing module**, where it is analyzed using predefined logic and calculations. The system then **generates the Digital Addiction Risk Score (DARS)** based on the processed data.

After processing, the data is either **stored in the database (DB)** for future reference or directly used to generate outputs. The system then sends the processed information to the **output module**, where results such as risk score, insights, and recommendations are displayed.

Finally, the **user receives the output** through the interface in the form of dashboards, alerts, and suggestions, completing the data flow cycle.

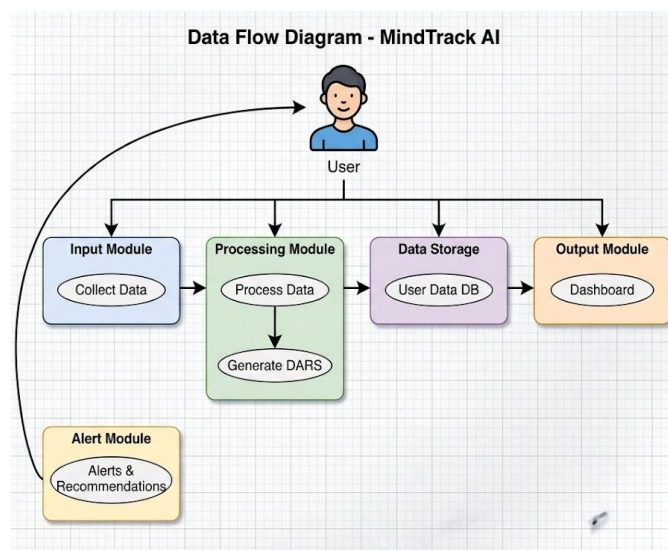


Fig 3.3:Data Flow Diagram

3.4 STATISTICAL ANALYSIS

The proposed system performs **Statistical Analysis (SA)** on user behavior data to identify meaningful patterns and trends related to digital usage. The analysis focuses on key parameters such as average screen time, app usage frequency, unlock count, and night-time activity. These parameters are collected continuously and used as inputs for evaluating user behavior.

Initially, the system calculates basic statistical measures such as **mean, frequency, and percentage** to understand the overall usage pattern. For example, average daily screen time is computed to determine how long a user spends on their device, while frequency analysis is used to measure how often the device is accessed. Night-time usage percentage helps in identifying unhealthy habits that may affect sleep patterns.

Based on these values, the system applies rule-based calculations to compute the **Digital Addiction Risk Score (DARS)**. The score is generated using weighted parameters, where each factor contributes differently to the final result. For instance, higher screen time and increased late-night usage result in a higher score, indicating greater risk. This approach simulates **Machine Learning (ML)**-like predictions without requiring complex models.

The system also uses **Data Analytics (DA)** techniques to compare current usage with previous data, enabling trend analysis. This helps in identifying whether user behavior is improving or worsening over time. The results of this analysis are displayed through charts and visual representations in the **User Interface (UI)**, making it easier for users to interpret their data.

Additionally, threshold-based analysis is used to trigger alerts through the **Real-Time Alert System (RTAS)**. When certain limits, such as excessive screen time or high night usage, are exceeded, the system generates notifications to warn the user. This ensures timely intervention and promotes better digital habits.

Overall, the statistical analysis component plays a crucial role in transforming raw data into meaningful insights. It provides a simple yet effective way to understand user behavior, generate risk scores, and support decision-making. Even without a fully implemented ML model, the system effectively demonstrates intelligent behavior through structured statistical methods.

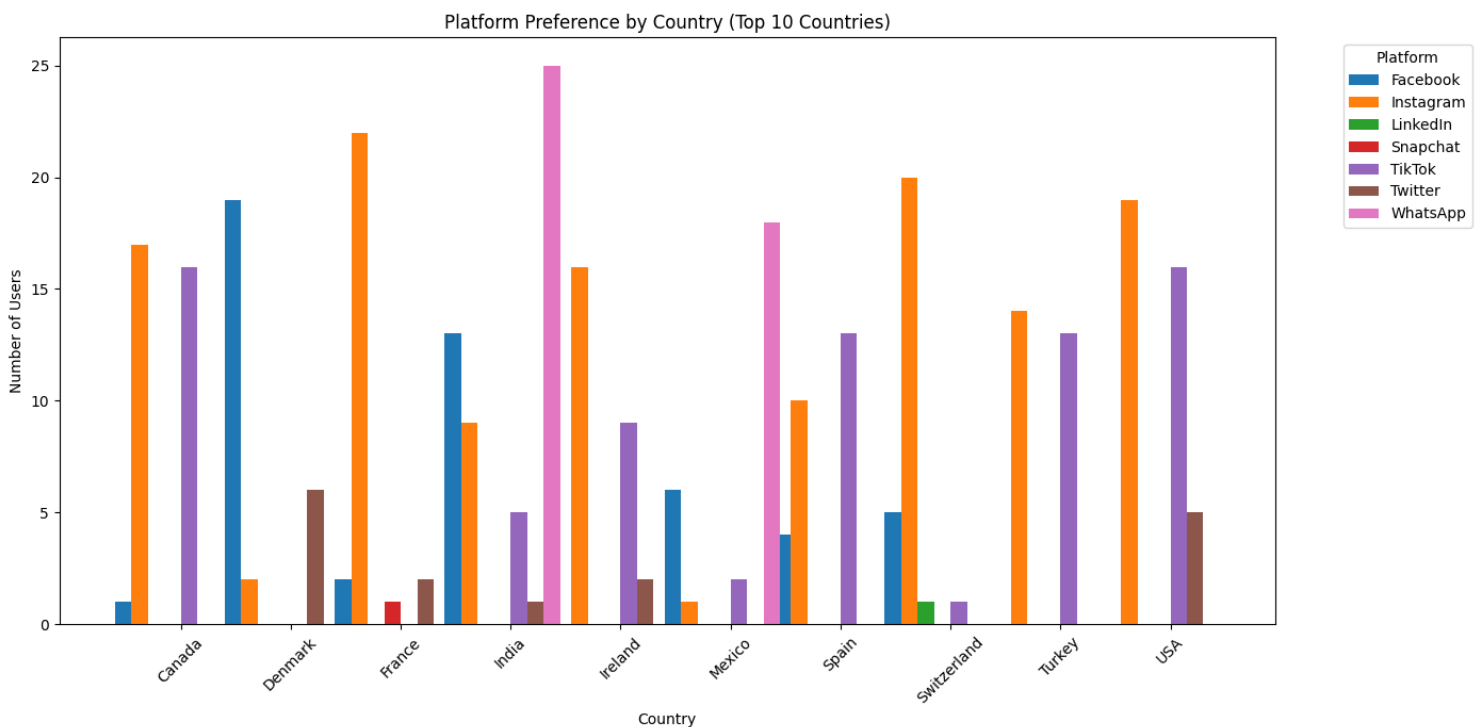


Fig 3.4 : Comparison Graph

CHAPTER 4

MODULE DESCRIPTION

The workflow for the proposed system is designed to ensure a structured and efficient process for analyzing user digital behavior and identifying potential addiction patterns. It consists of a series of sequential steps, including data collection, preprocessing, behavior analysis, risk score calculation, and result generation through insights, alerts, and recommendations.

4.1 SYSTEM ARCHITECTURE

4.1.1 USER INTERFACE DESIGN

The UI is designed using HTML, CSS, and JS to provide an interactive and user-friendly experience. It includes dashboards, charts, chatbot interface, and alert notifications. The design focuses on simplicity, responsiveness, and clear data visualization to enhance User Experience (UX).

4.1.2 BACK END INFRASTRUCTURE

The backend is responsible for handling data processing and overall system logic. It manages modules such as data handling, **Digital Addiction Risk Score (DARS)** calculation, and recommendation generation based on user behavior. The backend processes inputs like screen time, app usage, and unlock frequency to produce meaningful outputs. Instead of using full **Machine Learning (ML)** integration, it relies on lightweight rule-based logic, ensuring faster execution, reduced complexity, and easy deployment of the system.

4.2 DATA COLLECTION AND PREPROCESSING

4.2.1 Dataset and Data Labelling

The dataset consists of user behavior parameters such as screen time, usage frequency, and night activity. Labels such as Low Risk, Moderate Risk, and High Risk are assigned based on predefined thresholds. This helps in simulating classification similar to ML models.

4.2.2. Data Preprocessing

The dataset includes user behavior attributes such as screen time, usage frequency, unlock count, and night activity. Preprocessing involves cleaning the data, handling missing values, and applying normalization to ensure consistency. Threshold-based labeling is used to classify data into **Low, Moderate, and High Risk** categories. This step prepares structured data for analysis and simulates **Machine Learning** based classification.

4.2.3 Feature Selection

Feature selection is an important step in the system that focuses on identifying the most relevant parameters influencing digital addiction prediction. The dataset contains multiple behavioral attributes, but only significant features such as screen time, unlock count, social media usage duration, app switching frequency, and night-time activity are selected for analysis. These features are chosen based on their strong impact on user behavior patterns and addiction tendencies.

The selected parameters are further processed and used as primary inputs for calculating the **Digital Addiction Risk Score (DARS)**. Removing unnecessary or redundant features helps reduce computational complexity, improves processing speed, and increases the overall efficiency of the system. Proper feature selection also enhances the reliability and accuracy of behavior analysis by focusing only on high-impact user activity metrics.

4.2.4 CLASSIFICATION AND MODEL SELECTION

Multiple classification approaches were conceptually considered, including **Logistic Regression (LR)** for binary classification, **Random Forest (RF)** for handling multi-feature data, and **Gradient Boosting (GB)** for improved prediction accuracy. However, due to the lightweight nature of the system, a **rule-based classification model** was selected as the final approach.

This model classifies users into categories such as **Low, Moderate, and High Risk** based on threshold values derived from key behavioral features like screen time, unlock frequency, and night usage. The approach ensures reduced computational complexity while effectively simulating **Machine Learning (ML)**-like decision-making.

4.2.5 PERFORMANCE EVALUATION AND OPTIMIZATION

The performance of the system is evaluated using consistency-based metrics such as **prediction reliability, response time, and logical accuracy**. Although traditional ML evaluation metrics like **accuracy, precision, recall, and confusion matrix** are considered conceptually, the system primarily focuses on stable and interpretable outputs.

The rule-based model is iteratively refined by adjusting threshold values and feature weights to improve classification quality and minimize incorrect risk categorization. This optimization process ensures better alignment between user behavior patterns and generated results, enhancing overall system effectiveness.

4.2.6 MODEL DEPLOYMENT

The system is deployed as a web application using the **Python Flask** framework. It can run in both local and server environments, allowing users to access the application through a web browser. Since the system uses lightweight rule-based logic instead of complex **Machine Learning** models, deployment becomes faster, efficient, and easy to maintain.

4.2.7 CENTRALIZED SERVER AND DATABASE

A centralized server and database are used to store user activity details and processed results such as the **Digital Addiction Risk Score (DARS)**. The server manages communication between the frontend and backend, ensuring smooth data flow and real-time updates. This centralized structure supports efficient data management, quick retrieval, and scalability for handling multiple users.

4.3 SYSTEM WORK FLOW

This module describes the overall working process of the system from input to output generation. It ensures smooth flow of data and interaction between different modules.

4.3.1 USER INTERACTION

The user interacts with the system through the UI by providing or generating usage data. The system captures this data for analysis. The interface also displays results and insights.

4.3.2 BEHAVIOR ANALYSIS

The system analyzes user data to identify patterns and trends in digital behavior by examining parameters such as screen time, app usage frequency, unlock count, and night-time activity. It processes these inputs using predefined rules and Data Analytics techniques to detect irregular or excessive usage patterns. Based on these parameters, the system calculates the Digital Addiction Risk Score by assigning appropriate weights to each factor and combining them into a final score.

This score is then used to classify users into different risk levels, providing a clear indication of their digital dependency. By continuously evaluating and updating this score, the system enables better understanding of user behavior and supports informed decision-making for improving digital habits.

4.3.3 CHATBOT INTERACTION

An interactive chatbot assists users by answering queries and providing suggestions. It enhances engagement and provides a conversational experience. The chatbot uses predefined logic for responses.

4.3.4 ALERT & RECOMMENDATION SYSTEM

The system generates alerts when unhealthy usage patterns are detected. It also provides personalized recommendations to improve digital habits. This helps users take corrective actions.

4.3.5 CONTINUOUS MONITORING AND IMPROVEMENT

The system continuously monitors user activity by tracking parameters such as screen time, app usage, and unlock frequency in real time. This data is dynamically processed to update the **Digital Addiction Risk Score** and related insights without delay. Based on changes in user behavior, the system adapts its recommendations and alert mechanisms accordingly. This continuous monitoring and adaptive response help users gradually improve their digital habits and maintain better control over their usage patterns.

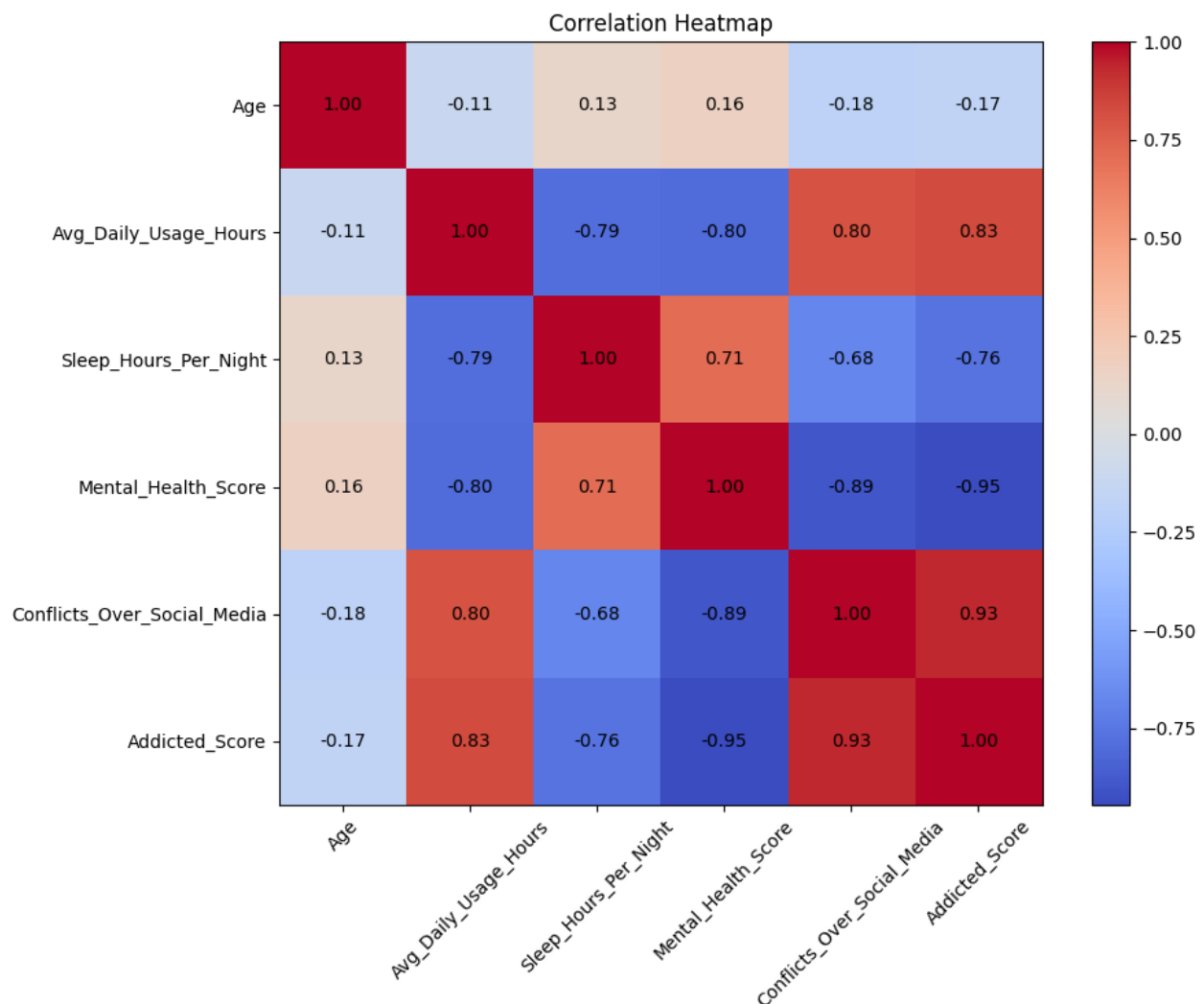


Fig :4.1 Correlation Heatmap

CHAPTER 5

IMPLEMENTATION AND RESULTS

5.1 IMPLEMENTATION

The proposed system, MindTrack AI, is implemented as a web-based application using modern frontend technologies such as .HTML, CSS, and JS. The UI is designed to be interactive, responsive, and user-friendly, enabling users to easily navigate through different features such as dashboards, chatbot, and alerts. The design follows a modular approach, where each component is structured separately for better maintainability and scalability. The core logic of the system is implemented using JavaScript, which handles data processing, behavior analysis, and Digital Addiction Risk Score calculation. User inputs such as screen time, app usage, and unlock count are processed using predefined rule-based algorithms. These algorithms simulate Machine Learning behavior by assigning weights to different parameters and generating a final risk score. This approach ensures efficient execution without the need for complex ML model integration.

For data visualization, libraries such as Chart.js are used to represent user behavior through line charts, bar graphs, and pie charts. These visual elements help users clearly understand their usage patterns and trends over time. The system also includes a chatbot module, implemented using JavaScript, which provides predefined responses based on user queries. This enhances Human-Computer Interaction and improves user engagement. Additionally, a Real-Time Alert System is implemented to notify users when unhealthy usage patterns are detected, such as excessive screen time or high night-time activity. The system dynamically updates data and results without requiring page reloads, ensuring a smooth and real-time experience. Overall, the implementation focuses on simplicity, efficiency, and effective demonstration of intelligent behavior using lightweight technologies.

5.2 OUTPUT SCREENSHOTS

The output screenshots section provides a visual representation of the working system and its various features. These screenshots demonstrate how the application processes user data and presents results through an interactive interface. Each screenshot highlights a specific functionality of the system, making it easier to understand the overall workflow and performance.

The **dashboard view** displays key metrics such as screen time, app usage, and unlock count in the form of cards and charts. It provides a comprehensive overview of user activity along with graphical representations for better interpretation. The **risk score display** shows the calculated **Digital Addiction Risk Score (DARS)** using a visual indicator such as a progress bar or circular chart, along with classification into low, moderate, or high risk.

The **chatbot interface** screenshot illustrates how users can interact with the system by asking questions and receiving instant responses. It showcases the conversational aspect of the application and highlights its ability to provide guidance and suggestions. The **alert notification screen** demonstrates how the system generates warnings when unhealthy usage patterns are detected, such as prolonged screen time or excessive night usage.

These screenshots collectively validate the functionality and effectiveness of the system. They provide clear evidence of how the application operates in real time, from data input to result generation. This section is essential for demonstrating the practical implementation and user interaction aspects of the project.

```

=== Most_Used_Platform ===
count      705
unique      12
top         Instagram
freq        249
Name: Most_Used_Platform, dtype: object
['Instagram' 'Twitter' 'TikTok' 'YouTube' 'Facebook' 'LinkedIn' 'Snapchat'
 'LINE' 'KakaoTalk' 'VKontakte' 'WhatsApp' 'WeChat']
=== Affects_Academic_Performance ===
count      705
unique       2
top         Yes
freq        453
Name: Affects_Academic_Performance, dtype: object
['Yes' 'No']
=== Relationship_Status ===
count      705
unique       3
top         Single
freq        384
Name: Relationship_Status, dtype: object
['In Relationship' 'Single' 'Complicated']

```

Fig 5.1 Dataset for Training

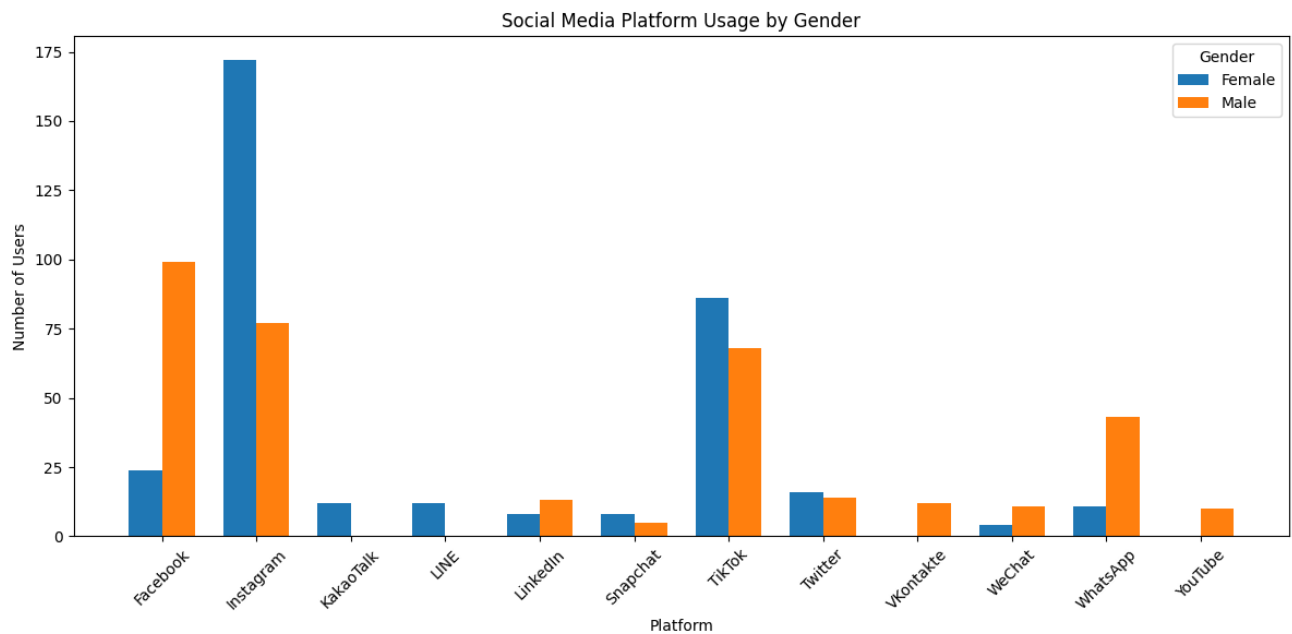


Fig 5.2 Performance Evaluation & Optimization

...	precision	recall	f1-score	support
2	0.00	0.00	0.00	1
3	0.67	1.00	0.80	2
4	1.00	1.00	1.00	15
5	1.00	1.00	1.00	35
6	1.00	0.89	0.94	9
7	0.98	1.00	0.99	41
8	1.00	1.00	1.00	28
9	1.00	1.00	1.00	10
accuracy			0.99	141
macro avg	0.83	0.86	0.84	141
weighted avg	0.98	0.99	0.98	141
[[0 1 0 0 0 0 0 0 0]				
[0 2 0 0 0 0 0 0 0]				
[0 0 15 0 0 0 0 0 0]				
[0 0 0 35 0 0 0 0 0]				
[0 0 0 0 8 1 0 0 0]				
[0 0 0 0 0 41 0 0 0]				
[0 0 0 0 0 0 28 0 0]				
[0 0 0 0 0 0 0 10 0]				

Fig 5.3 Confusion Matrix

	Model	train_mae	train_mse	train_rmse	train_r2	test_mae	test_mse	test_rmse	test_r2
0	Random Forest	0.017402	0.004220	0.064960	0.998324	0.037908	0.024449	0.156363	0.990229
1	Xgboost	0.077933	0.019010	0.137878	0.992448	0.096005	0.033903	0.184128	0.986451
2	Linear Regression	0.141322	0.044482	0.210907	0.982330	0.221484	0.104504	0.323271	0.958237

Fig 5.4 Best Algorithm

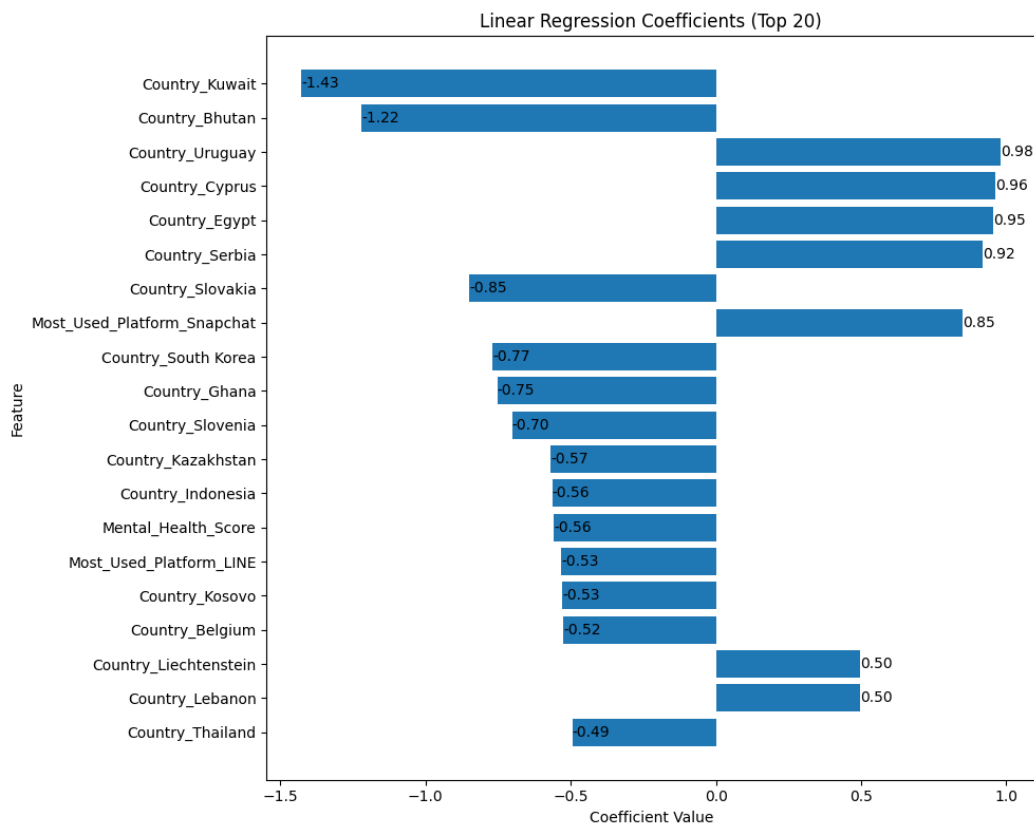


Fig 5.5 Linear Regression Coefficients



Fig 5.6 Webpage for Addiction score, analysis and reminder

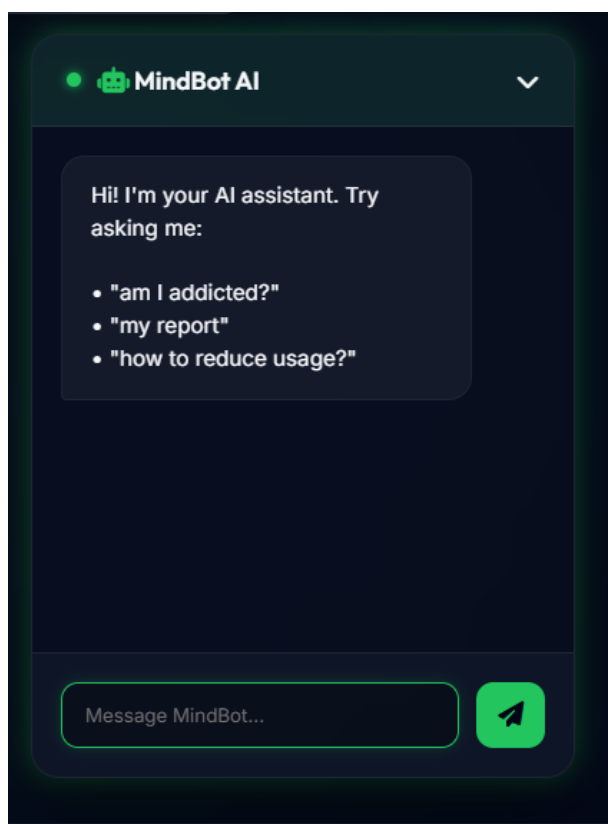


Fig 5.7 Chatbot

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

The **MindTrack AI** system successfully demonstrates an effective approach to analyzing user digital behavior through a web-based platform developed using **Python Flask**. It collects and processes user activity data such as screen time, app usage, and unlock frequency to generate a **Digital Addiction Risk Score (DARS)**, enabling users to clearly understand their digital habits and usage patterns. The system simulates concepts of **Artificial Intelligence (AI)** and **Machine Learning (ML)** using rule-based logic, which ensures simplicity, efficiency, and ease of implementation while still providing meaningful insights. With integrated features such as real-time analytics, interactive chatbot support, and a **Real-Time Alert System (RTAS)**, the application actively engages users and provides timely feedback on unhealthy usage patterns. Overall, the system serves as a practical tool for promoting digital awareness, improving productivity, and encouraging users to adopt healthier and more balanced technology usage habits.

6.2 FUTURE ENHANCEMENT

The **MindTrack AI** system can be further enhanced by integrating advanced technologies to improve accuracy and user experience. In the future, real **Machine Learning (ML)** models can be used for more precise predictions and deeper insights. Developing a mobile application can improve accessibility and real-time monitoring. Additionally, advanced **Data Analytics (DA)** and cloud deployment can enhance scalability and performance. Features like a voice-enabled chatbot and improved **User Experience (UX)** can make the system more interactive and effective.

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